

Blender 3D Made Super Easy!

Think of Blender like a massive digital toy box filled with virtual Play-Doh, colorful stickers, flashlights, and a movie camera!

1. What Even is Blender? (The Playground)

Imagine you have a magic sandbox inside your computer screen. Inside this sandbox, you can build anything you want out of digital clay, paint it with bright colors, shine flashlights on it, and record a video of it moving around!

🎮 MOVING YOUR HEAD AROUND THE TOY BOX

To look around your 3D world, use your mouse buttons like magic eyes:

- **Look Around (Rotate):** Click and hold the **Middle Mouse Wheel** and drag your mouse around.
- **Slide Sideways (Pan):** Hold **Shift** + hold **Middle Mouse Wheel** and drag.
- **Get Closer / Farther (Zoom):** Scroll your **Mouse Wheel** up and down.

2. The 3 Magic Keys (Moving Toys Around)

When you put a toy cube in your world, you need to move it, spin it, or make it huge or tiny. You only need to remember **3 Magic Letters** on your keyboard:

1. The "G" Key (Grab)

Press **G** to **Grab** your object and slide it around the screen. Press **X**, **Y**, or **Z** right after to move it in a straight line!

2. The "R" Key (Rotate)

Press **R** to **Rotate** or spin your object around like a toy top.

3. The "S" Key (Scale)

Press **S** to **Scale** it! Drag mouse away to make it giant, or close to make it tiny.

3. Playing with Digital Play-Doh (Edit Mode)

Right now, your block is just a solid block. But what if you want to squish it or pull out nose and ears? You need to enter **Edit Mode**!

MAGIC PORTAL: THE TAB KEY

Press the **Tab** key on your keyboard to switch between **Object Mode** (holding the whole toy) and **Edit Mode** (squishing the Play-Doh).

Super Simple Shaping Tools

TOOL NAME	SHORTCUT	HOW TO THINK ABOUT IT
Pinch Points	1 , 2 , 3	Choose to grab individual dots (vertices), edges, or full flat sides (faces).
Extrude (Pull Out)	E	Grabs a flat side and pulls it out like pulling cheese off a hot pizza slice!
Loop Cut (Slice)	Ctrl + R	Acts like a laser knife to slice your clay in half so you have more parts to bend.
Bevel (Smooth Edges)	Ctrl + B	Sands down sharp dangerous corners so they look smooth and soft.

4. Painting & Stickers (Materials)

Everything in 3D starts out plain grey like cardboard. We need to paint it and give it textures!

- **Base Color:** Pick any color in the rainbow (red, blue, neon green!).
- **Roughness:**
 - Set to **0** = Super shiny like a clean glass mirror or shiny sports car.
 - Set to **1** = Dull and rough like chalk or paper.
- **Metallic:** Turn this up to **1** to make your object look like real gold, silver, or robot steel!

5. Flashlights & The Camera

You can't see anything in a dark room without flashlights. In Blender, lights work just like real room lamps!

1. **Main Light (Key Light):** A big bright light pointing at your toy's face.
2. **Soft Fill Light:** A dimmer light on the side so dark shadows aren't too scary pitch black.
3. **Back Light (Rim Light):** A light behind your toy to make its edges glow like a superhero!

6. Making it Move! (Keyframe Animation)

Animation is just taking pictures of a toy, moving it slightly, and taking another picture!



THE 3-STEP ANIMATION TRICK

1. Set your timeline at **Frame 1**. Place your toy, press **I**, and pick *Location*. (Blender remembers where it started!).
2. Move the timeline to **Frame 60** (2 seconds later). Move your toy to a new spot.
3. Press **I** again to save the new spot. Hit **Spacebar** and watch your toy walk on its own!

7. Making Your Final Movie (Rendering)

When you are done, click **Render** → **Render Animation**! Blender will take all your frames, light them up, and save a real **.mp4** video file onto your computer that you can share on TikTok, YouTube, or with friends!